THE GREAT INDIAN AI INTERNSHIP CHALLENGE — GNANI AI

AgriMind AI – Agentic Smart Agriculture Assistant

Voice-First Multilingual Agricultural Decision Support for Indian Farmers

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GitHub Repository: https://github.com/Karthik-9465/agrimind.git

PROBLEM STATEMENT

The Challenge for Indian Farmers: Smallholder and regional farmers in India face significant barriers when accessing timely, actionable agricultural intelligence. Text-heavy advisory portals, complex English agronomic jargon, and low digital literacy create severe adoption bottlenecks.

The Interaction Gap: Conventional agritech requires manual typing and English navigation. When crop leaves develop lesions or weather threats emerge, farmers need immediate advice spoken in their native mother tongue—without deciphering bureaucratic spreadsheets or navigating complex apps.

PROPOSED SOLUTION

AgriMind AI Ecosystem: A unified voice-first assistant that enables farmers to simply speak in their mother tongue, photograph plant foliage, and receive real-time spoken and visual agronomic decision support tailored to their specific land parcels.

Verified Capabilities: (1) Regional voice interaction (STT/TTS), (2) Crop leaf health diagnosis via vision, (3) FAO-56 Penman irrigation scheduling, (4) ICAR STCR soil-test fertilizer planning, (5) Hyperlocal agro-weather & spray window advisories, (6) APMC mandi wholesale price trends, and (7) Multilingual conversation in Telugu, Hindi, and Indian English.

CORE INNOVATION: GNANI AI VOICE SUITE

Voice-First Interaction Architecture: Transforms complex agronomy into natural conversational speech powered by official Gnani AI endpoints.

- **Gnani Prisma STT (v3):** Captures farmer spoken audio, applies 16kHz WAV normalisation, and transcribes voice into text with high Indic regional accuracy.
- **Gnani Timbre TTS (v3.0):** Converts synthesized agronomic advice into crystal-clear 48kHz audio streams, enabling farmers to listen effortlessly.
- **Voice Pipeline Flow:** *Farmer speaks → Prisma transcribes → AgriMind processes query → Timbre generates spoken response.*
- **Verified Regional Language Matrix:** Telugu (te-IN with voice *Avinash*), Hindi (hi-IN with voice *Shivani*), and Indian English (en-IN with voice *Avinash*). Dynamic speed clamping (0.85x–1.15x) ensures natural, comfortable listening.

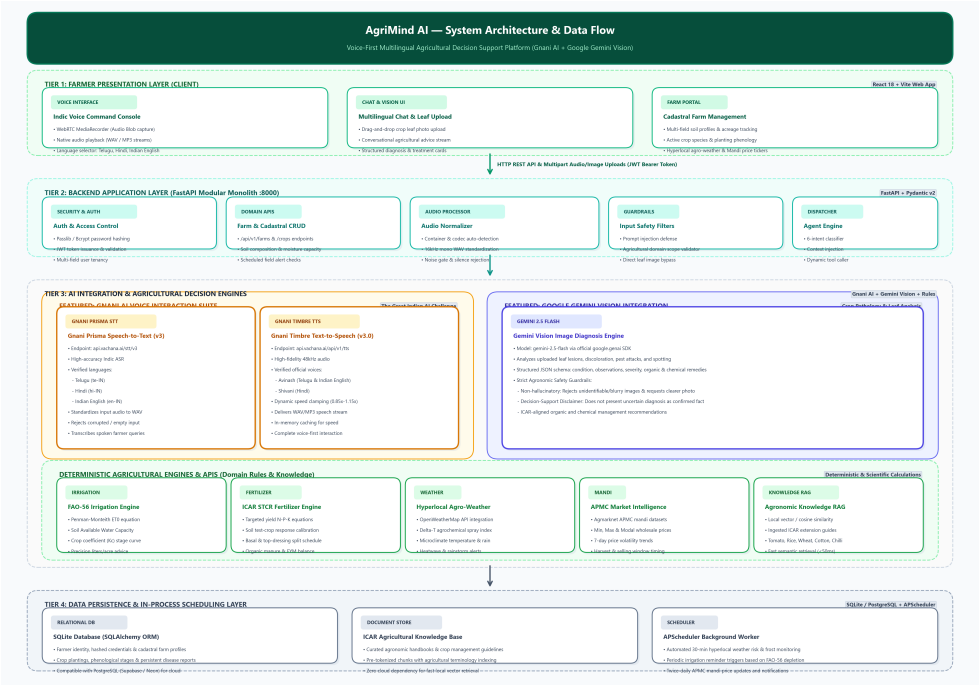
CROP HEALTH: GOOGLE GEMINI 2.5 FLASH VISION

Multimodal Plant Pathology & Diagnosis: Evaluates uploaded leaf photographs to detect foliar lesions, blight, pest damage, and nutritional deficiencies.

- **Structured Diagnostics:** Extracts condition, severity, observable symptoms, and ICAR-aligned organic and chemical management remedies.
- **Strict Agronomic Safety Guardrails:** Non-hallucinatory evaluation rejects unidentifiable or blurry photos and instructs the farmer to upload a clearer image.
- **Decision Support, Not Guaranteed Diagnosis:** All analyses explicitly carry an agronomic advisory disclaimer and recommend field expert confirmation for critical outbreaks.
- **Model:** Official gemini-2.5-flash with 60s in-memory caching and a dual-attempt automated network retry loop.

TECHNICAL SYSTEM ARCHITECTURE & INTEGRATED DATA FLOW

Modular Monolith Architecture: React 18 + Vite Frontend (Voice WebRTC recorder & Cadastral Farm Portal) → **FastAPI Monolith (:8000)** with JWT Auth & Input Guardrails → **Agentic Router** dispatching to **Gnani Prisma STT / Timbre TTS**, **Gemini Vision**, and 5 deterministic scientific engines (FAO-56 Irrigation, ICAR STCR Fertilizer, Hyperlocal Weather, APMC Mandi, and Agronomy RAG) → **SQLite (SQLAlchemy) & APScheduler**.



IMPACT, SCIENTIFIC RIGOR & COMPETITIVE INNOVATION

1. Regional Voice Inclusion

- Eliminates text literacy and keyboard friction by enabling direct voice conversation in Telugu, Hindi, and Indian English.
- Bridges the digital divide for smallholder farmers who prefer listening over reading complex advisory brochures.
- End-to-end hands-free interaction designed specifically for field conditions.

2. Deterministic Agronomy

- Zero LLM hallucinations on critical farm dosages: Uses **FAO-56 Penman-Monteith** equations for millimeter-precise irrigation schedules.
- Employs official **ICAR STCR targeted yield equations** for N-P-K fertilizer balancing and soil protection.
- Incorporates real-time **Delta-T spray window indexes** to prevent pesticide evaporation or drift.

3. Economic Empowerment

- Real-time APMC Mandi price discovery protects growers from distress sales and middleman exploitation.
- Local vector RAG indexed across ICAR agronomy handbooks delivers instantaneous extension advisory.
- Clean modular monolith architecture with 100% automated test coverage (31/31 passed) and production readiness.